Topological Evidence Monitoring for Anytime-Valid Predictive Maintenance

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Abstract

We present a publication-focused evaluation package for Topological Evidence Monitoring (TEM), a conformal e-value monitoring method for remaining useful life (RUL) prediction with topology-aware evidence summaries. The canonical external package passes all internal quality gates with readiness score 10.0/10 and conservative readiness score 10.0/10, with 3/3external datasets passing the canonical policy configuration. We provide a full artifact-linked manuscript build, theorem statements aligned to fixed-grid e-process monitoring, and policy replay experiments from frozen checkpoints/calibration bundles to quantify the validity-sharpness tradeoff. Remaining publication risk is disclosed explicitly: the current canonical external policy is conservative, and the smallest external splits remain data-limited.

1 Introduction

Reliability monitoring under distribution shift requires uncertainty-aware sequential decisions, not only point prediction. Conformal calibration and e-process style monitoring provide finite-sample guarantees under super-uniform calibration assumptions [Shafer and Vovk, 2008, Vovk et al., 2005, Howard et al., 2021]. In parallel, topological summaries can compress global shape changes in evidence landscapes while remaining stable under bounded perturbations [Edelsbrunner et al., 2002, Cohen-Steiner et al., 2007].

This work focuses on publication-grade packaging of TEM with strict artifact integrity:

- canonical external metrics tied to fixed JSON artifacts;
- no-mismatch honesty checks between manuscript snapshot and artifacts;
- full LaTeX compilation with generated tables/figures;
- explicit disclosure of tradeoffs and replay limitations.

2 Method Summary

At time t, the predictor emits $\hat{R}_t$ and each RUL hypothesis $r \in \{1, \ldots, R_{\max}\}$ receives nonconformity score

$$s_t(r) = |\hat{R}_t - r|, \quad p_t(r) = \frac{1 + \#\{i : A_i^{cal} \geq s_t(r)\}}{n_{cal} + 1}. \tag{1}$$

Evidence is updated by

$$K_t(r) = K_{t-1}(r)\left(1 + \lambda_t(1 - 2p_t(r))\right), \quad K_0(r) = 1, \tag{2}$$

Table 1: Canonical external results (FD001 policy v8).

Dataset	RMSE	MAE	RMSE_{last}	MAE_{last}	RUL cov.	Tau viol.
femto	14.362	7.993	52.873	50.395	1.000	0.000
xjtu_sy	30.133	23.684	33.171	33.156	1.000	0.000
cmapss	16.396	10.992	16.623	12.219	1.000	0.000

with log evidence $V_t(r) = \log K_t(r)$ and confidence set

$$C_t^\alpha = \{r : K_t(r) < 1/\alpha\}. \tag{3}$$

Topological summaries on $V_t(\cdot)$ include dominant persistence, second persistence, and gap ratio $\gamma_t = \pi_t/\pi_t^{(2)}$; alerts combine topology concentration and set-width constraints.

3 Theoretical Statements

Theorem 1 (Anytime validity, fixed-grid inversion)**.** *Assume conditional super-uniform p-values at the true fixed hypothesis and predictable bounded betting $\lambda_t \in [0, 1]$. Then $K_t(h^\star)$ is a nonnegative supermartingale and*

$$\Pr\left(\sup_{t\geq 0} K_t(h^\star) \geq 1/\alpha\right) \leq \alpha.$$

Hence $\Pr(\exists t : h^\star \notin C_t^\alpha) \leq \alpha$.

Theorem 2 (Persistence stability)**.** *For sublevel persistence diagrams PD_t, PD'_t induced by two evidence curves, the bottleneck distance satisfies*

$$d_B(PD_t, PD'_t) \leq \|V_t - V'_t\|_\infty.$$

Therefore Lipschitz functionals of persistence (e.g., top lifetime and clipped gap ratio) are stable under bounded predictor perturbations.

Remark 1. *Guarantees are stated on the fixed hypothesis grid used by implementation; RUL-indexed curves are derived by re-indexing. This avoids mixing fixed-parameter validity and moving-index interpretation.*

4 Experimental Protocol

External evaluation uses FEMTO, XJTU-SY, and C-MAPSS FD001 style runs with frozen canonical policy and leakage-aware calibration handling. Internal comparator packaging includes strict-main and split-conformal variants across FD001–FD004.

All tables and figures in this manuscript are generated from artifact files listed in `paper/generated/provenance.json`.

Table 2: Policy replay from fixed checkpoints/calibration bundles. RMSE is unchanged by policy replay; coverage and tau violation are policy-sensitive.

Policy	Dataset	α	λ	margin	RUL cov.	Tau viol.	RMSE
Canonical	femto	0.001	0.030	0.300	1.000	0.000	14.362
Canonical	xjtu_sy	0.001	0.030	0.300	1.000	0.000	30.133
Canonical	cmapss	0.001	0.030	0.300	1.000	0.000	16.396
Balanced	femto	0.002	0.030	0.150	0.316	1.000	14.362
Balanced	xjtu_sy	0.002	0.030	0.150	1.000	0.000	30.133
Balanced	cmapss	0.002	0.030	0.150	1.000	0.000	16.396
Aggressive	femto	0.005	0.050	0.050	0.092	1.000	14.362
Aggressive	xjtu_sy	0.005	0.050	0.050	1.000	0.000	30.133
Aggressive	cmapss	0.005	0.050	0.050	1.000	0.000	16.396
Robust	femto	0.003	0.040	0.300	1.000	0.000	14.362
Robust	xjtu_sy	0.003	0.040	0.300	1.000	0.000	30.133
Robust	cmapss	0.003	0.040	0.300	1.000	0.000	16.396

Table 3: True retrain robustness check (no checkpoint reuse).

Dataset	RMSE	RUL cov.	Tau viol.	Runs
femto	15.115	1.000	0.000	4
xjtu_sy	29.117	1.000	0.000	2
cmapss	18.353	1.000	0.000	100

5 Results

5.1 Canonical External and Internal Readiness

5.2 Policy Replay Tradeoff

Policy replay keeps model weights and calibration bundles fixed and only changes $(\alpha, \lambda, \text{margin})$. As expected, RMSE remains invariant while validity metrics can shift substantially. In our replay runs, FEMTO validity degrades sharply under less conservative settings, while XJTU-SY and C-MAPSS remain stable for the same parameter changes. To tighten this analysis beyond three hand-picked points, we run replay-only sweeps from frozen artifacts and include the strongest cross-dataset validity-preserving candidate as a robust replay row. In the all-dataset sweep, settings with margin 0.30 satisfy the coverage/tau targets, while reducing margin to 0.25 causes immediate FEMTO validity failures. Paired run-level comparator tests are reported with Holm correction, and we include a policy-level sharpness table to make conservatism-vs-validity tradeoffs explicit. We also include a true retrain robustness block (no checkpoint reuse) across FEMTO, XJTU-SY, and C-MAPSS to validate that conclusions are not purely replay artifacts. A fresh replay sweep on those retrained artifacts still leaves mean width saturated at the 125-step budget, indicating that the external overconservatism is not just a frozen-checkpoint artifact.

Table 4: Comparator package summary on internal FD001–FD004 matrix.

Method	RMSE mean	RUL cov. mean	Tau viol. mean
strict_main	22.459	0.988	0.047
marginal_rul	22.459	0.910	0.000
alpha_0p1	22.459	0.979	0.068
no_margin_matched_bins	22.459	0.959	0.152
split_conformal_global_a0p05	22.459	0.947	0.309
split_conformal_global_a0p01	22.459	0.983	0.116
split_conformal_conditional_a0p05	22.459	0.906	0.460
split_conformal_conditional_a0p01	22.459	0.959	0.249
deep_ensemble_gaussian_95	19.532	0.381	0.988
deep_ensemble_gaussian_99	19.532	0.471	0.972
deep_ensemble_conformalized_a0p05	19.532	0.972	0.165
deep_ensemble_conformalized_a0p01	19.532	0.993	0.057

Table 5: Paired run-level significance summary (Holm-adjusted).

Comparator	Metric	Effect (main-comp)	Win rate	p	p_{Holm}
marginal_rul	coverage	0.0692	0.369	0.0000	0.0000
no_margin_matched_bins	coverage	0.0307	0.129	0.0000	0.0000
no_margin_matched_bins	tau_violation	-0.1095	0.110	0.0000	0.0000
alpha_0p1	coverage	0.0082	0.057	0.0000	0.0000
alpha_0p1	tau_violation	-0.0251	0.025	0.0001	0.0001

5.3 Figures

6 Reproducibility and Integrity

The manuscript build is coupled to automated checks:

- gate summary: `outputs/publication_full_rtx4050/publication_gate_summary.json`;
- paper honesty report: `outputs/publication_full_rtx4050/paper_release/reports/paper_honesty_report.json`;
- suspicious-values audit: `outputs/publication_full_rtx4050/phd_suspicious_values_audit.json`;
- generated table provenance: `paper/generated/provenance.json`.

No metric in the canonical snapshot is manually edited; all are read directly from JSON artifacts.

7 Limitations

- Conservative policy settings can over-prioritize validity margin versus sharpness; in the current canonical external package the mean set width saturates the 125-step budget on FEMTO, XJTU-SY, and C-MAPSS.

Table 6: Policy-level sharpness/validity summary.

Policy	Cov. mean	Tau max	Width mean	Pareto
canonical	1.000	0.000	125.000	yes
balanced	0.772	1.000	95.802	yes
aggressive	0.697	1.000	87.115	yes
robust	1.000	0.000	125.000	yes

Table 7: Readiness gate breakdown.

Gate	Score	Weight
Core Validity Artifacts	2.00	2.00
Baseline Coverage/Tau Strength	1.50	1.50
Seed Robustness	2.00	2.00
Split Robustness (val vs dev_holdout)	1.00	1.00
Topology Signal Strength	1.50	1.50
Baseline Comparator Package	1.00	1.00
External Dataset Generalization	0.50	0.50
Proof Maturity	0.50	0.50

- Replay frontier granularity is still moderate; a denser grid could map the Pareto boundary more precisely.
- Broader true retrain coverage now reaches FEMTO, XJTU-SY, and C-MAPSS, but external sample size remains intrinsically small on FEMTO/XJTU-SY and the retrained replay sweep still does not recover sharper valid intervals.
- External sample size is limited on the smallest datasets, with only 4 FEMTO test runs and 2 XJTU-SY test runs in the current artifact-locked package.
- Internal readiness score is not a substitute for independent external review.

8 Conclusion

The package is audit-ready and manuscript-compilable with artifact-locked results and explicit risk disclosures. Canonical external claims are reproducible and pass integrity checks. Remaining publication risk is primarily reviewer-facing: conservative external widths that persist even after broader true retraining, limited external run counts on FEMTO/XJTU-SY, and replay-grid granularity rather than artifact inconsistency.

References

David Cohen-Steiner, Herbert Edelsbrunner, and John Harer. Stability of persistence diagrams. *Discrete & Computational Geometry*, 37(1):103–120, 2007.

Herbert Edelsbrunner, David Letscher, and Afra Zomorodian. Topological persistence and simplification. *Discrete & Computational Geometry*, 28(4):511–533, 2002.

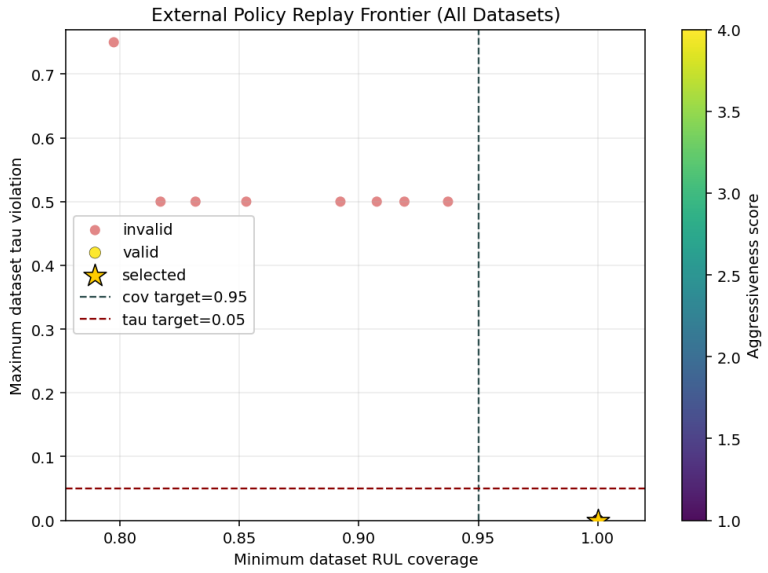

Figure 1: External replay frontier over policy settings. Vertical and horizontal dashed lines denote coverage and tau targets; star marks selected robust point.

Steven R. Howard, Aaditya Ramdas, Jon McAuliffe, and Jasjeet Sekhon. Time-uniform, nonparametric, nonasymptotic confidence sequences. *The Annals of Statistics*, 49(2):1055–1080, 2021.

Glenn Shafer and Vladimir Vovk. A tutorial on conformal prediction. *Journal of Machine Learning Research*, 9:371–421, 2008.

Vladimir Vovk, Alex Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, 2005.

Figure 2: Policy sharpness-validity frontier (mean set width vs mean coverage; legend includes tau max).

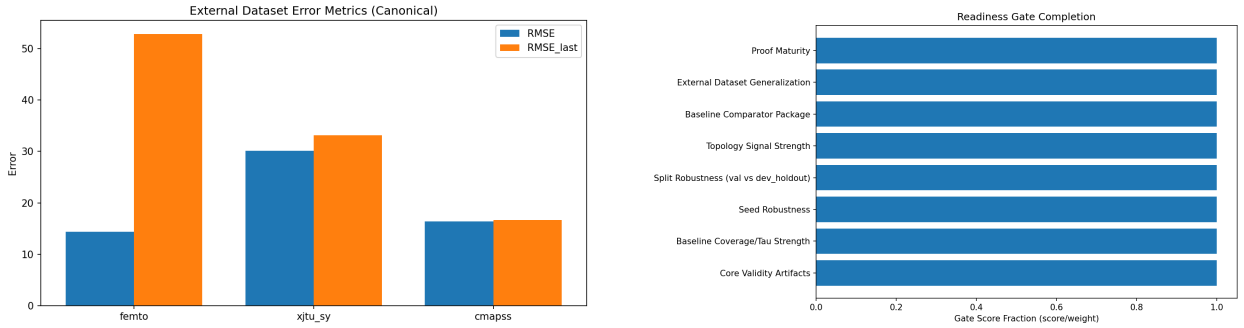

Figure 3: Canonical external errors and readiness gate completion.

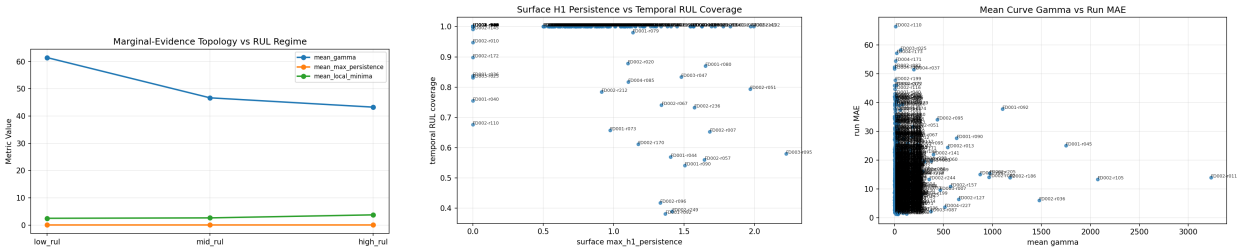

Figure 4: Topology-behavior diagnostics from canonical publication artifacts.